



Swethank Reddy Akketi

Bachelor of Technology
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PROFESSIONAL SUMMARY

IIT Bombay undergraduate with hands-on experience building deep learning systems, multi-agent LLM pipelines, and computer vision applications. Proficient in PyTorch, TensorFlow, LangChain, and LangGraph; achieved **3.2× inference speed-up** via GGUF quantization and ONNX optimization. Seeking a 2-3 month summer internship as an **AI/ML Intern** or **Software Engineering Intern** at an AI-first or deep tech company.

SCHOLASTIC ACHIEVEMENTS

- Ranked in the top **2.2%** among 180,000+ candidates in **JEE Advanced** 2023
- Ranked in the top **0.73%** among 1.1M+ candidates in **JEE Mains** 2023

KEY PROJECTS

Building Scalable Multi-Agent AI Systems (Summer 2025)
SOC Project | Web and Coding Club IIT Bombay

- Engineered **MLP, CNN, LSTM, Transformer** architectures in PyTorch with **94.3%** accuracy on MNIST and **89.1%** on IMDB sentiment analysis
- Built production-grade **RAG and ReAct agents** using **LangGraph** and **LangChain** with locally hosted LLMs
- Achieved **3.2× inference speed-up** (820 ms to 255 ms) via **GGUF quantization** and **ONNX** runtime optimization

Real-Time Object Detection & Facial Recognition (July 2024)
Self Project IIT Bombay

- Deployed **YOLOv8** for real-time multi-class object detection on live video streams
- Implemented **Siamese Neural Networks** for one-shot facial recognition with high inter-class discriminability

Conditional Image Synthesis with GANs (July 2024)
Self Project IIT Bombay

- Trained **Conditional GAN** in **Keras** to synthesize class-specific images; applied **data augmentation** and regularization to stabilize adversarial training

Gesture-Based Text Creation & Recognition (Summer 2024)
SOC Project | Web and Coding Club IIT Bombay

- Built **CNN** in **TensorFlow** for handwritten **MNIST** digit recognition with **89%+** accuracy
- Integrated **MediaPipe** hand-landmark detection with **OpenCV** for a real-time gesture-controlled drawing canvas

Market Regime Detection via Unsupervised Learning (April 2025)
Self Project IIT Bombay

- Applied **K-Means** clustering and **PCA** dimensionality reduction to order book data, identifying **11** distinct market regimes
- Achieved **Silhouette Score: 0.450** · **Davies-Bouldin Index: 1.227**

Bulldozer Price Prediction: Time Series ML (Summer 2024)
Self Project IIT Bombay

- Trained **Random Forest Regressor** on **400K+** record Kaggle dataset; achieved **R² = 0.88**
- Reduced overfitting via **RandomizedSearchCV** hyperparameter tuning

TECHNICAL SKILLS

Languages	Python, C++, JavaScript, HTML, CSS
ML / DL Frameworks	PyTorch, TensorFlow, Keras, Scikit-learn
AI / LLM Tools	LangChain, LangGraph, GGUF Quantization, ONNX, RAG, ReAct Agents
Computer Vision	OpenCV, MediaPipe, YOLOv8
Data Libraries	NumPy, Pandas
Web / Backend	ReactJS, Node.js, Django
Tools	GitHub, Arduino, AutoCAD, Blender, Adobe Suite

POSITIONS OF RESPONSIBILITY

Creative & Media Convener | Institute Technical Council, IIT Bombay (April'24–March'25)

- Grew social impressions by **115K+**; scaled outreach by **200%** through targeted digital campaigns

Events & PR Coordinator | Entrepreneurship-Cell, IIT Bombay (July'24–March'25)

- Managed *Eureka!* — Asia's largest Business Model Competition (INR 20M prize pool); onboarded **10+** incubators

College Representative Mentor | Abhyuday, IIT Bombay (August'24–February'25)

- Mentored **30+** students on academic and career planning

Creatives Coordinator | Inter IIT Tech Meet 13.0 (September'24–February'25)

- Achieved **5×** growth in social media reach through strategic content and visual design